

EEE 416 – Microprocessor and Embedded Systems Laboratory

Jan 2025 Level-4 Term-1 Section B2

Project Presentation

IoT-Based Smart Walking Stick for Enhanced Mobility of the Elderly and Visually Impaired People.



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Outline

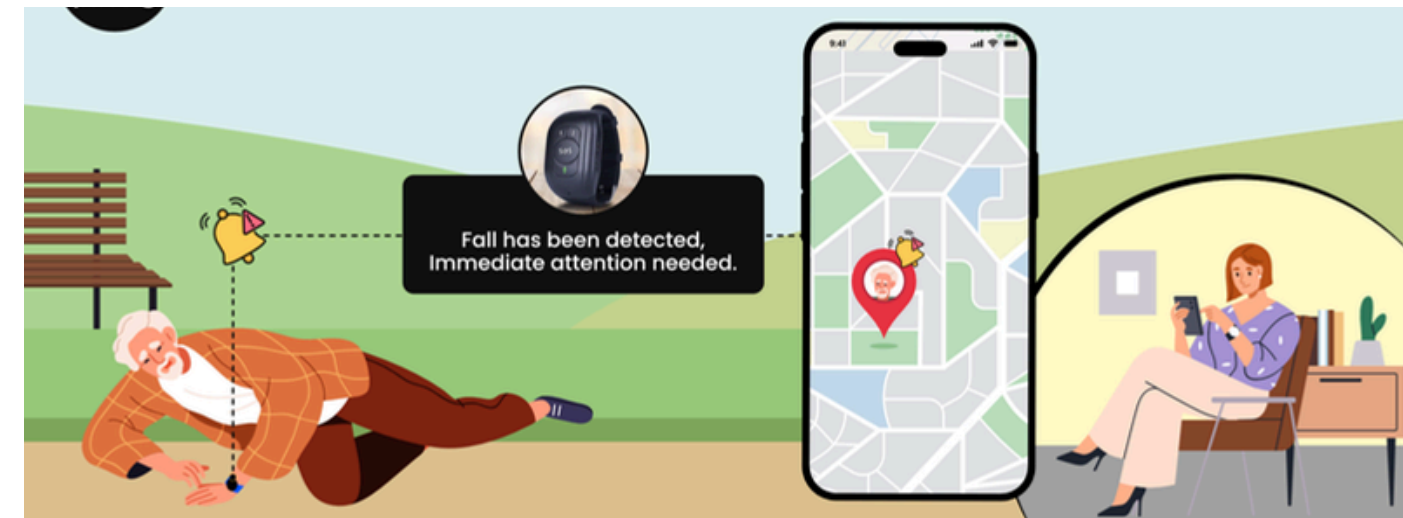
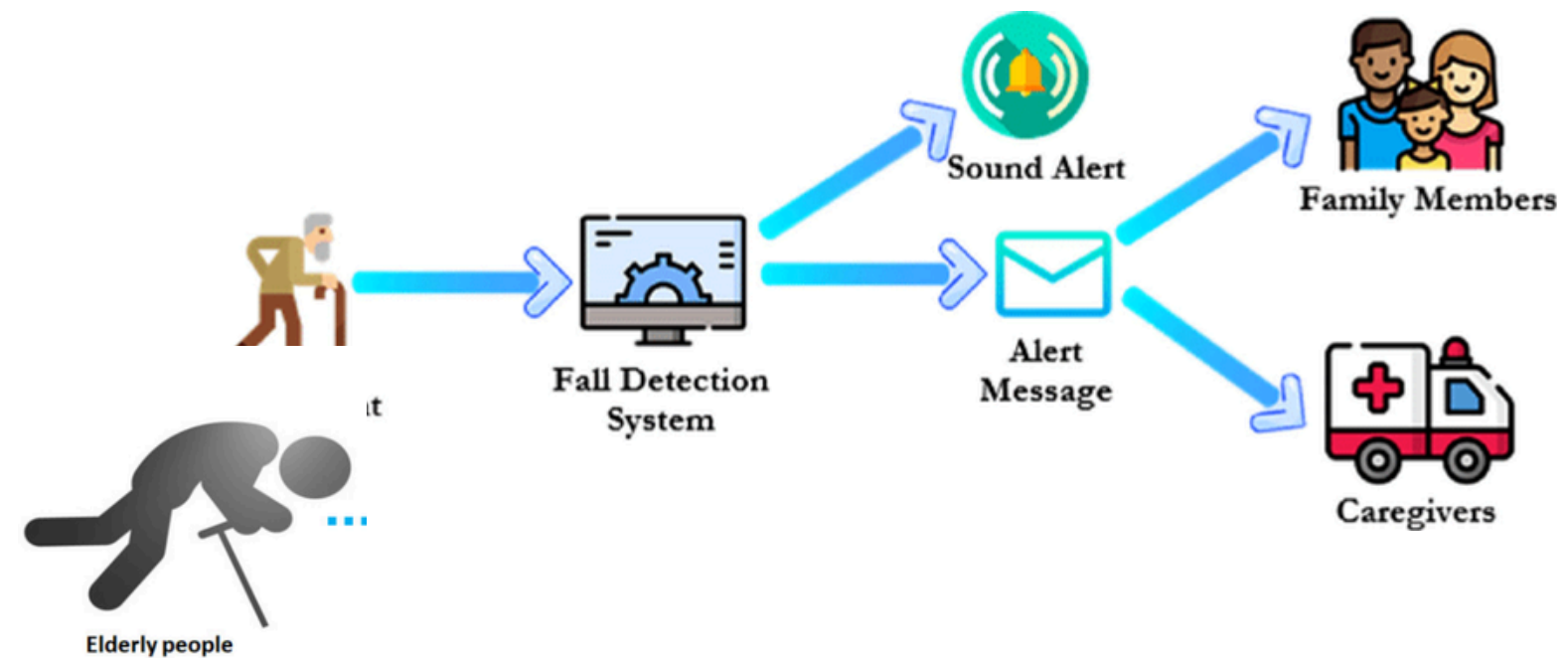
1. Background of the Project
2. System Architecture
3. sensor introduction
4. Detailed Sensor Integration
5. Object detection flowchart
6. Diagram of Methodology
7. GPS integration
8. Cost Analysis of Components
9. Division of Work
10. Timeline of project completion



Introduction

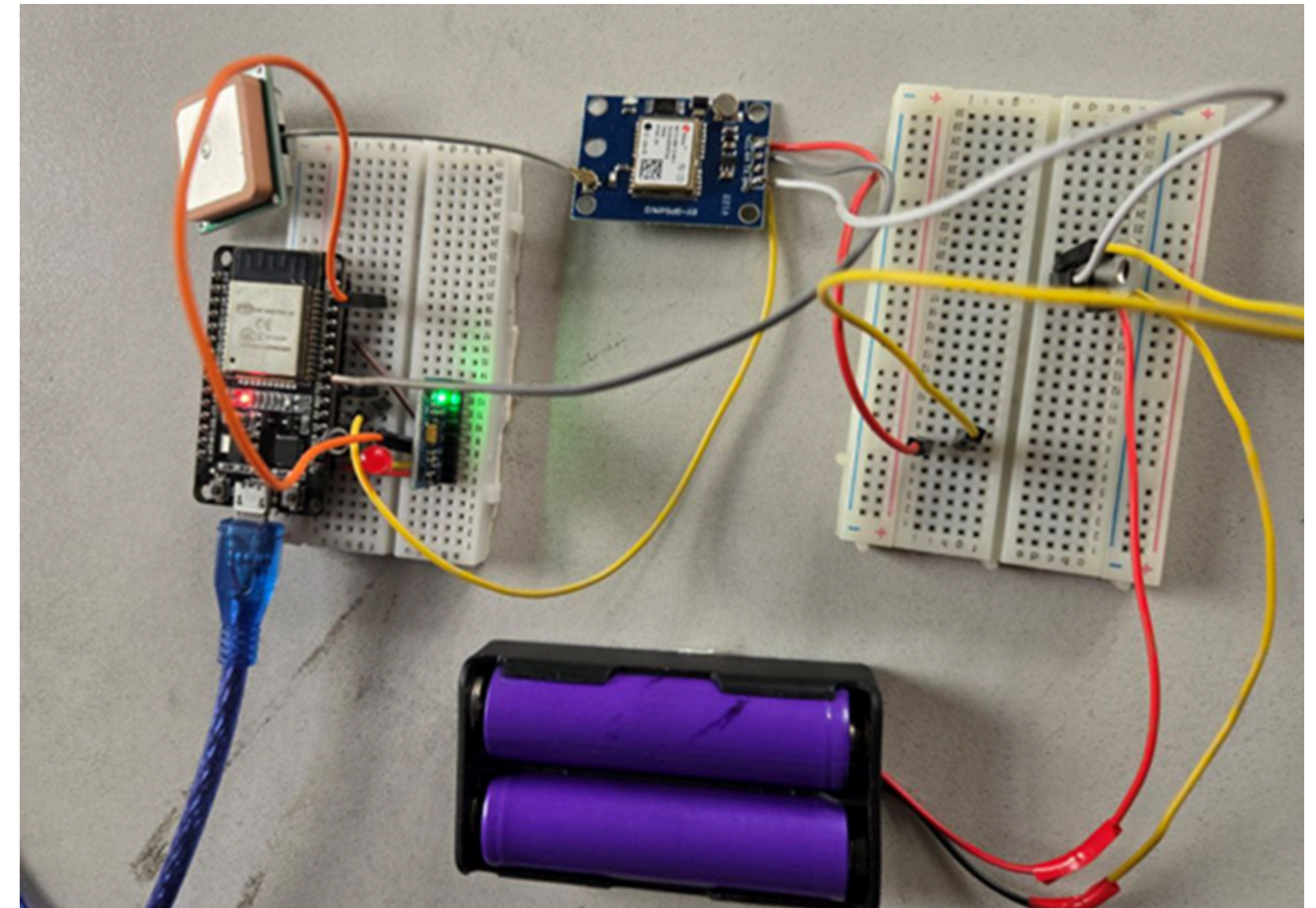
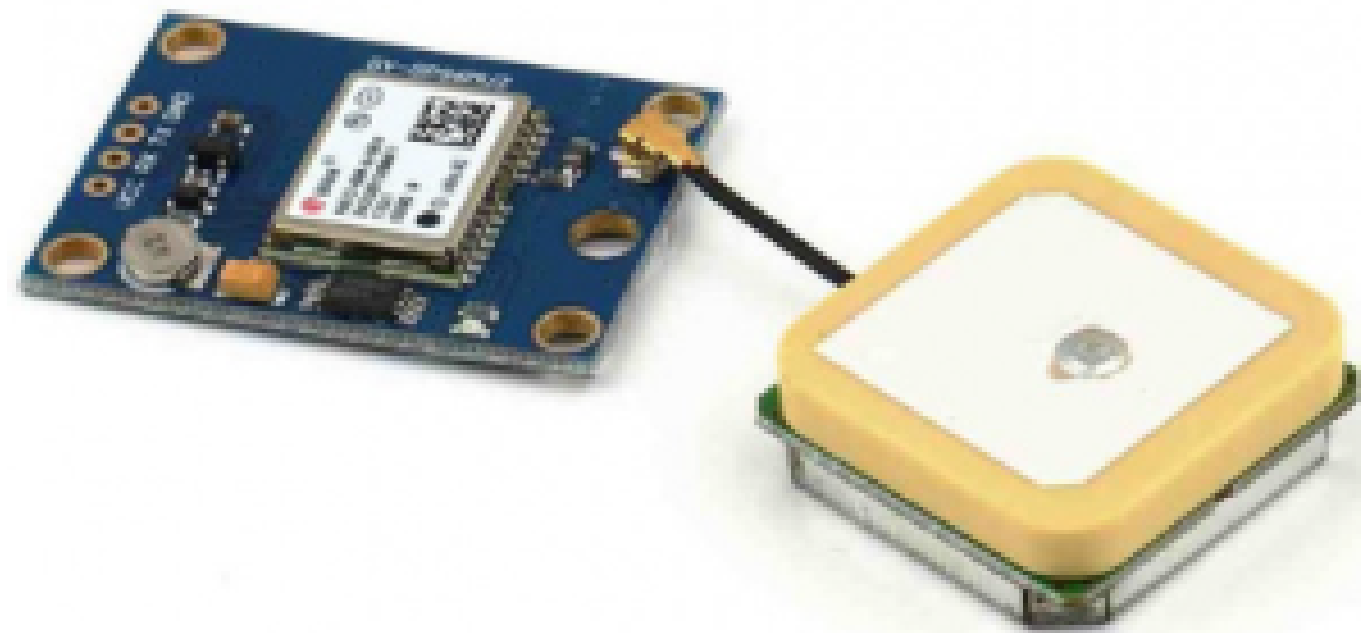
Key Features

- Fall Detection
- Obstacle Detection
- Location Tracking
- Web App Integration
- Alert and Notification System



Sensor Introduction

GPS Module: Used for real-time location tracking.



Sensor Introduction

Sensors: Detect obstacles in the path (front, left, right).



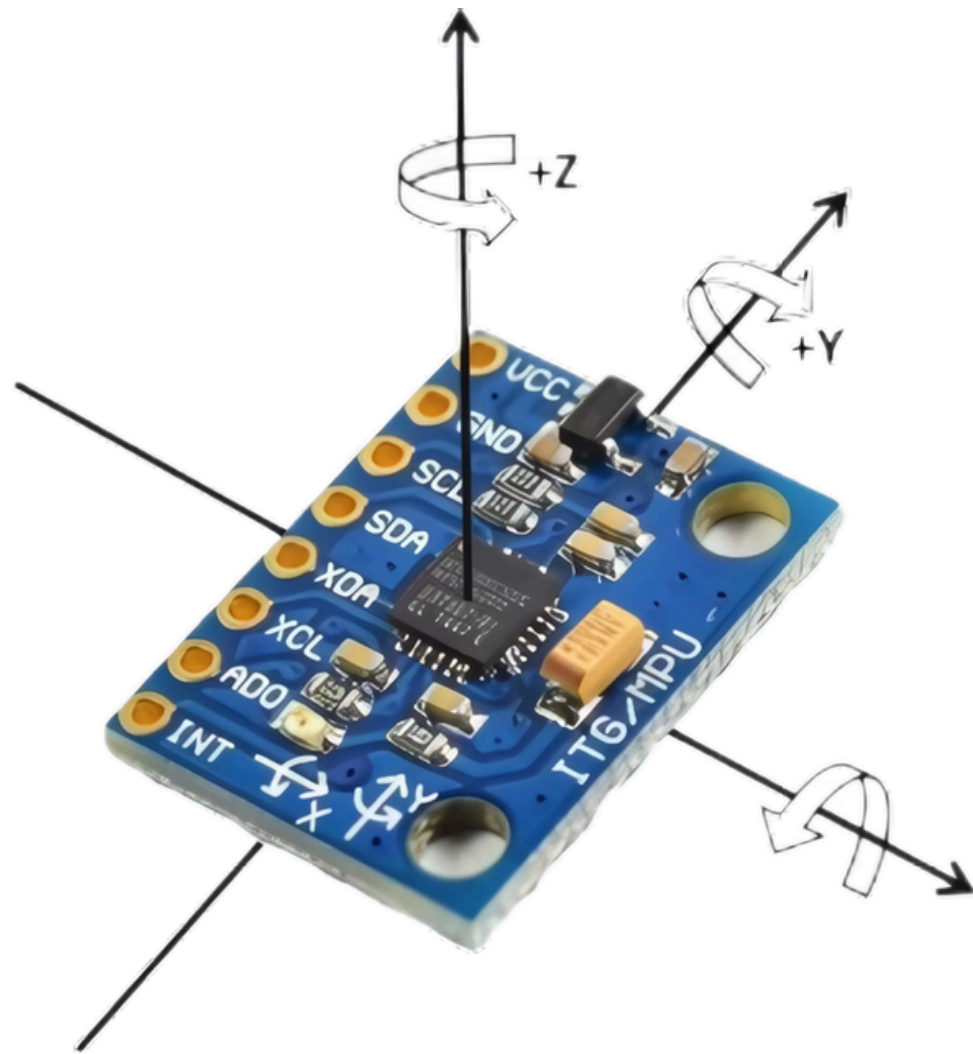
Sensor Introduction

water Sensors: Detect water in road.



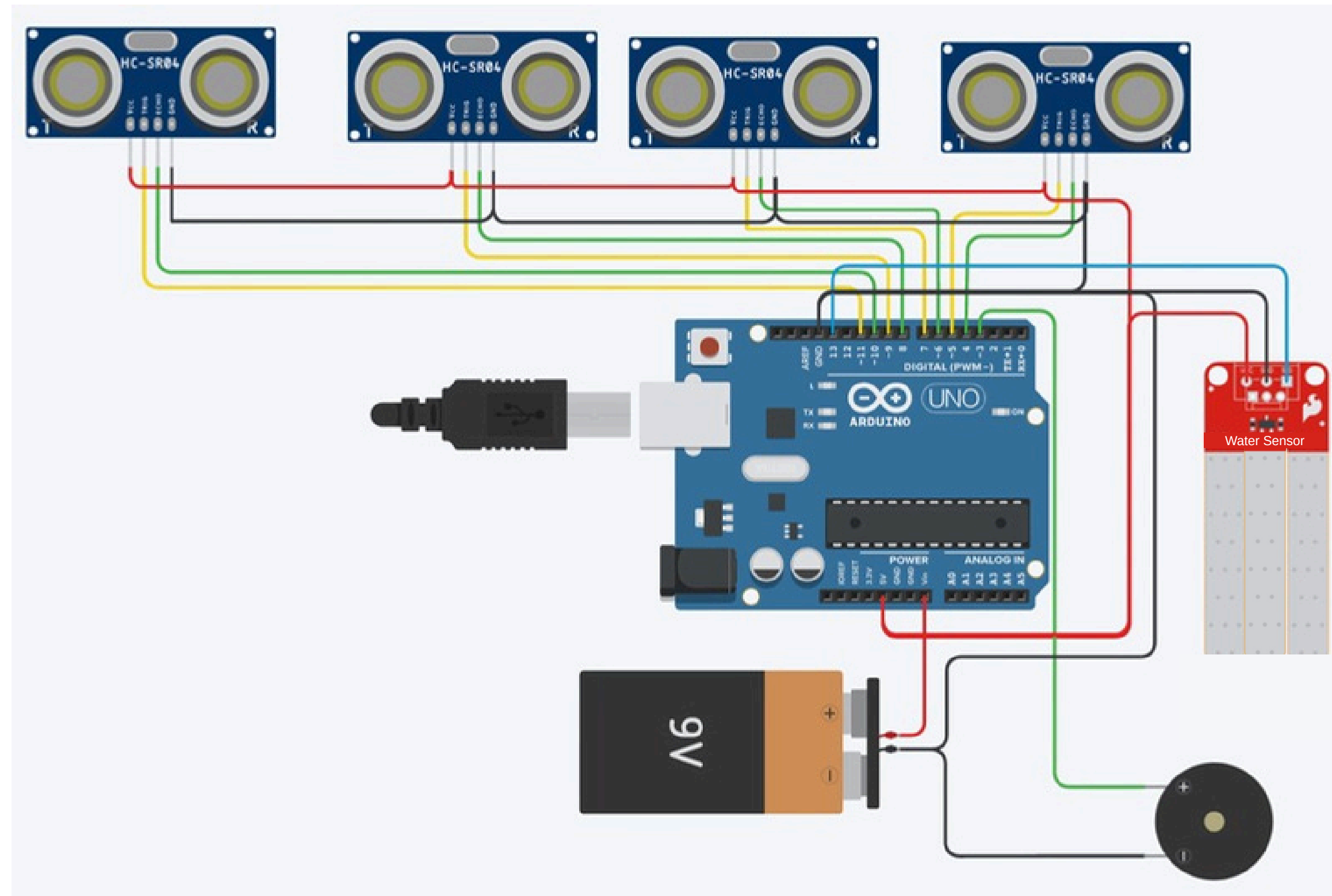
Detailed Sensor Introduction

Accelerometer: Used for fall detection and angle calculation.

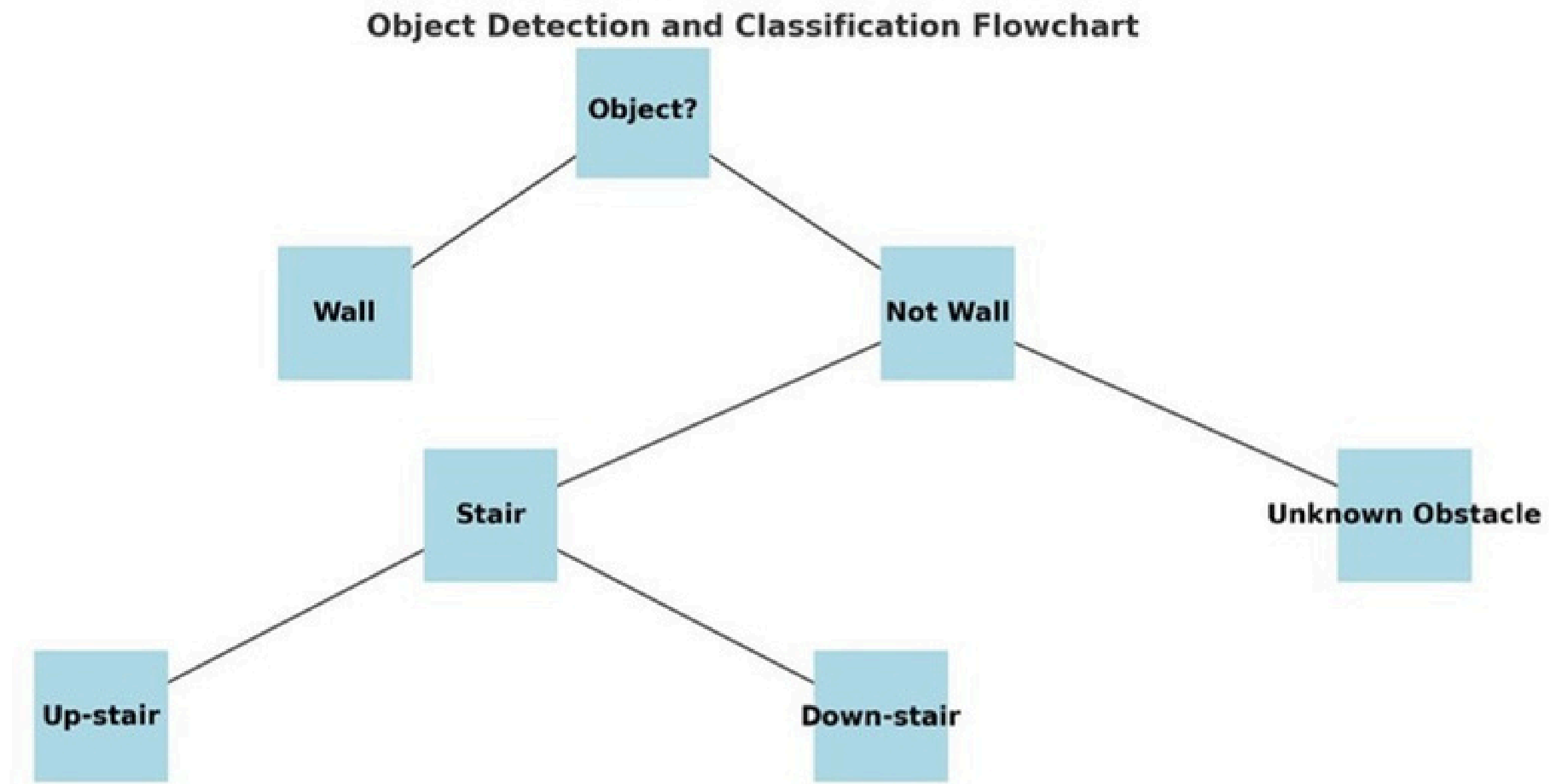


$$|a| = \sqrt{Ax^2 + Ay^2 + Az^2} \approx 1g \approx 9.8 \text{ m/s}^2$$

Sensor Integration

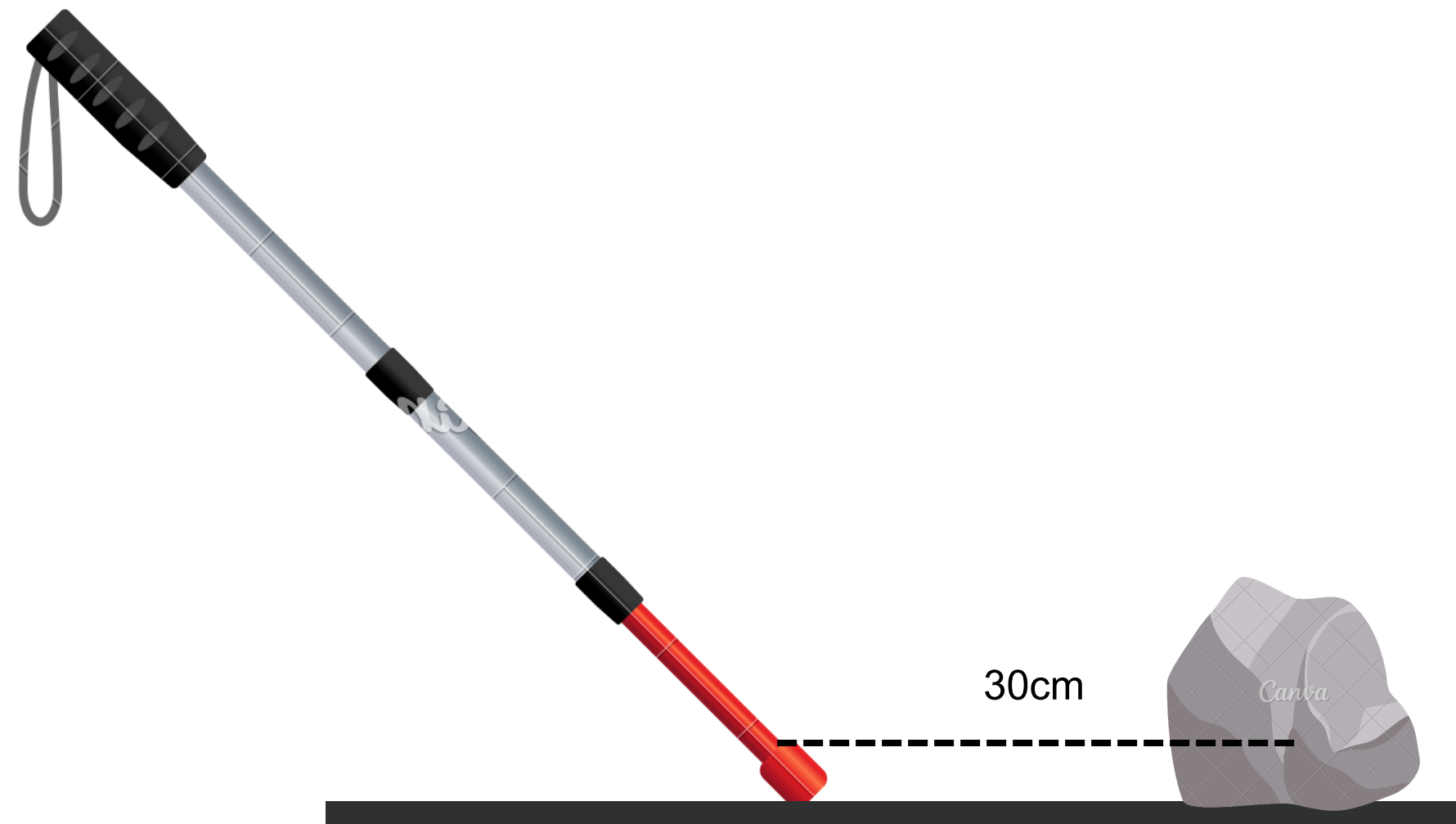


Object Detection and Classification Flowchart

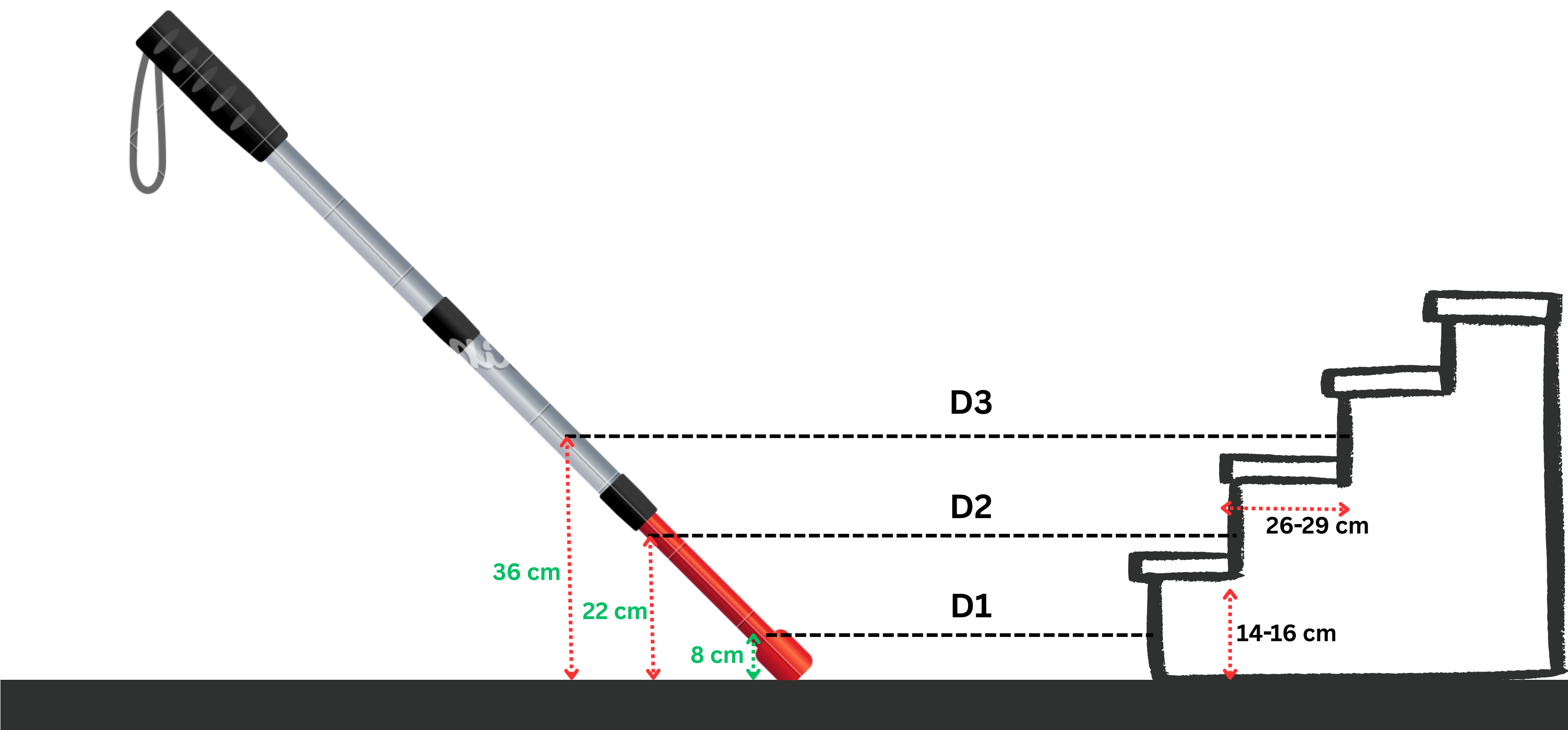


Object Detection

obstacle is detected
(any sensor < threshold)
Assumed Threshold = 30cm

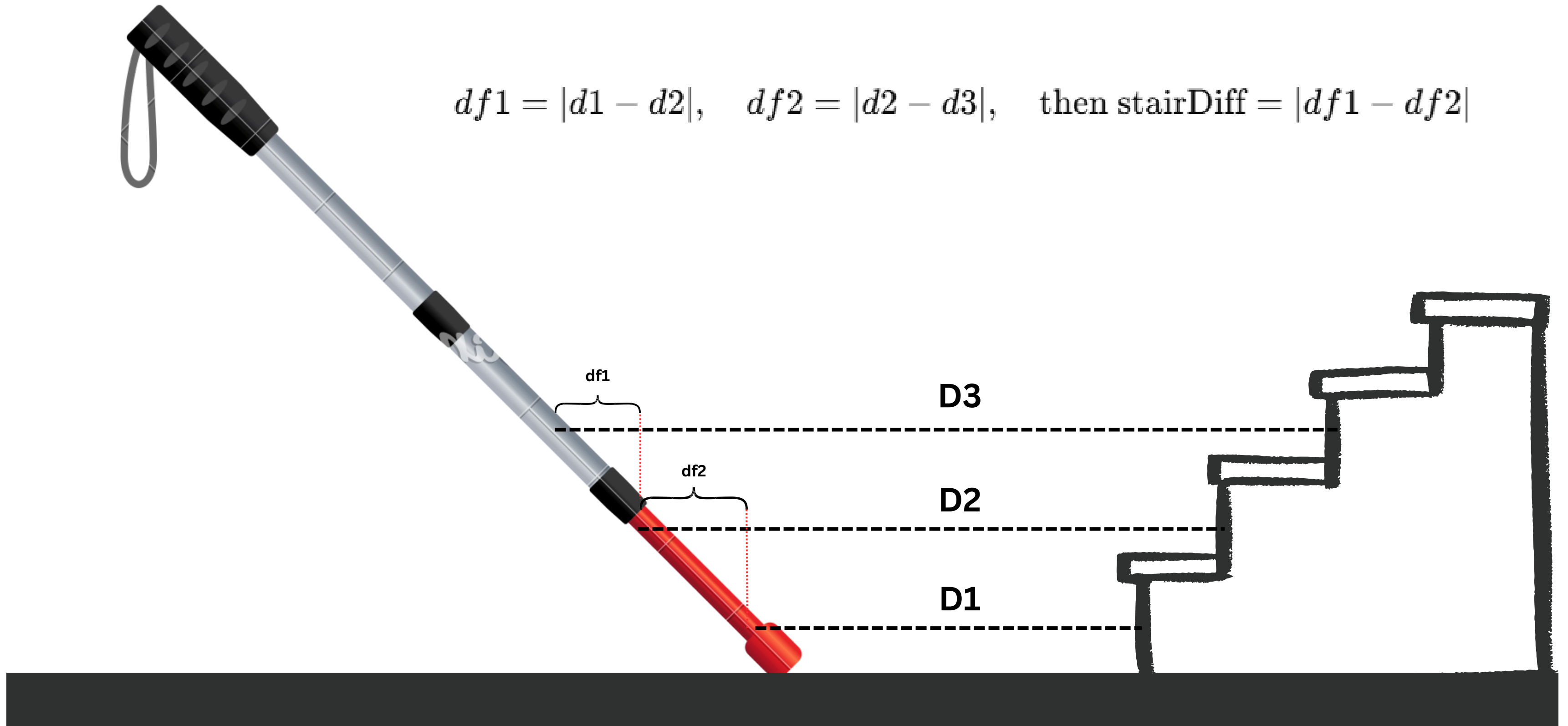


UPStair Detection

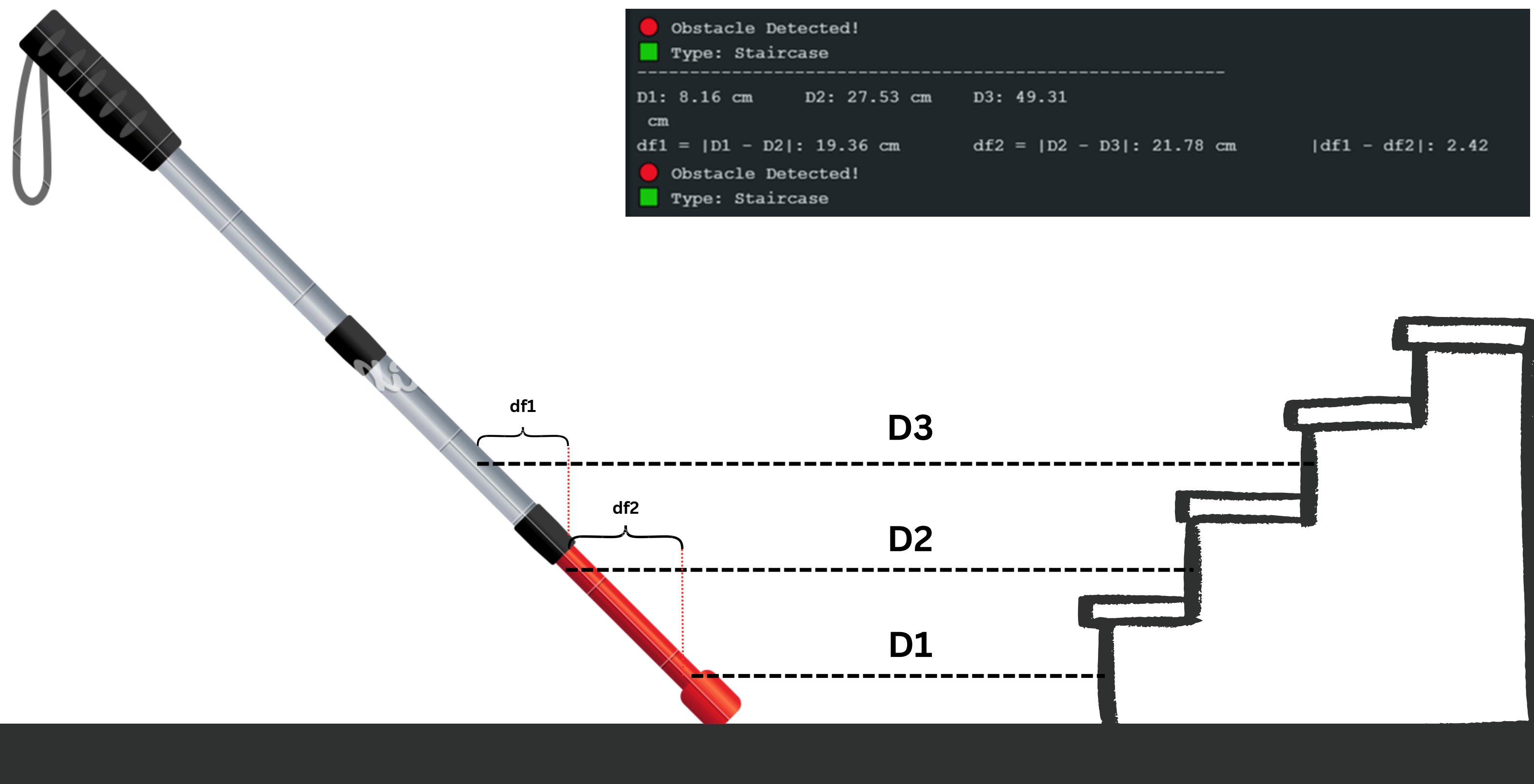


UPStair Detection

$$df1 = |d1 - d2|, \quad df2 = |d2 - d3|, \quad \text{then stairDiff} = |df1 - df2|$$

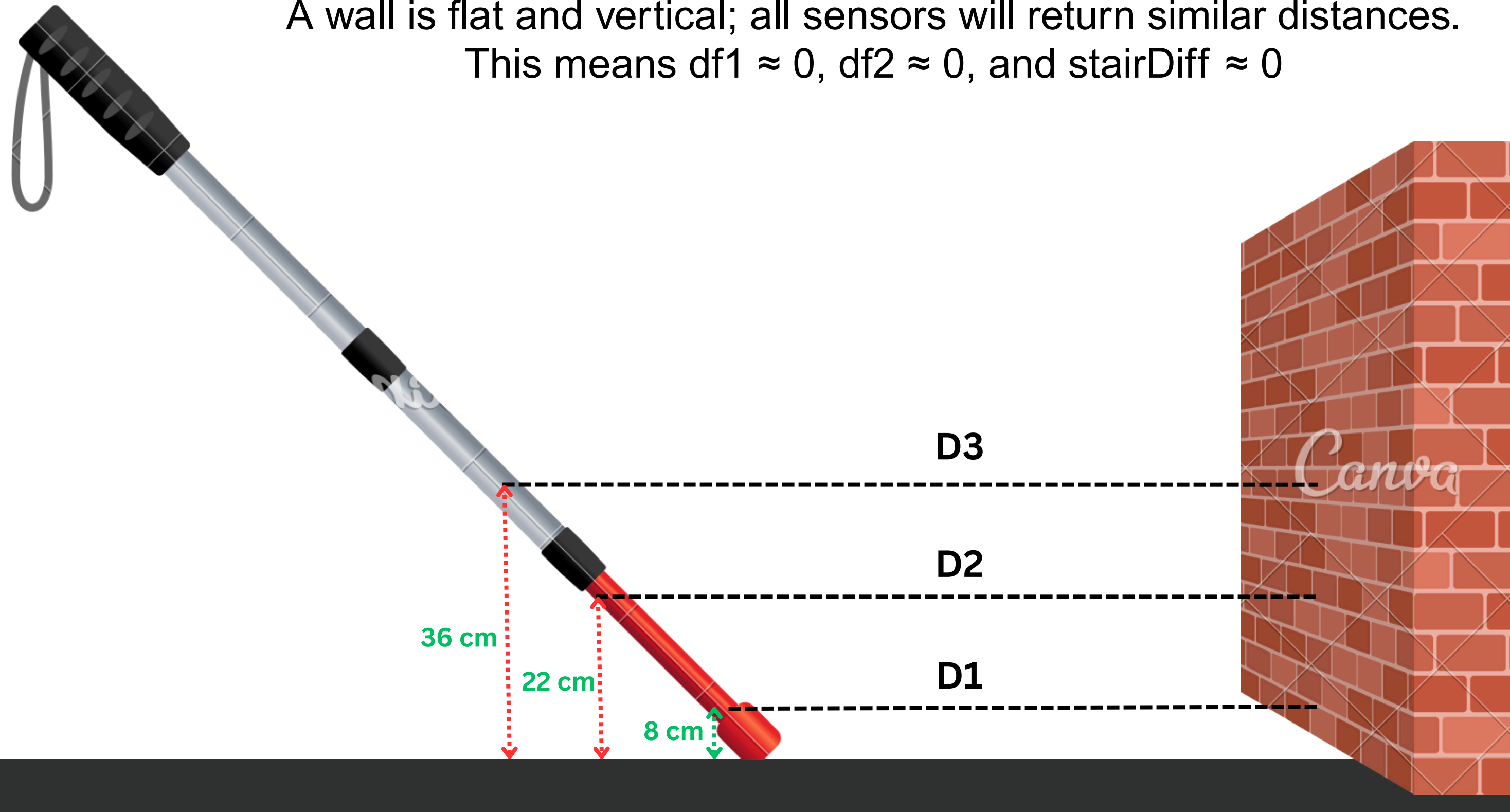


UPStair Detection

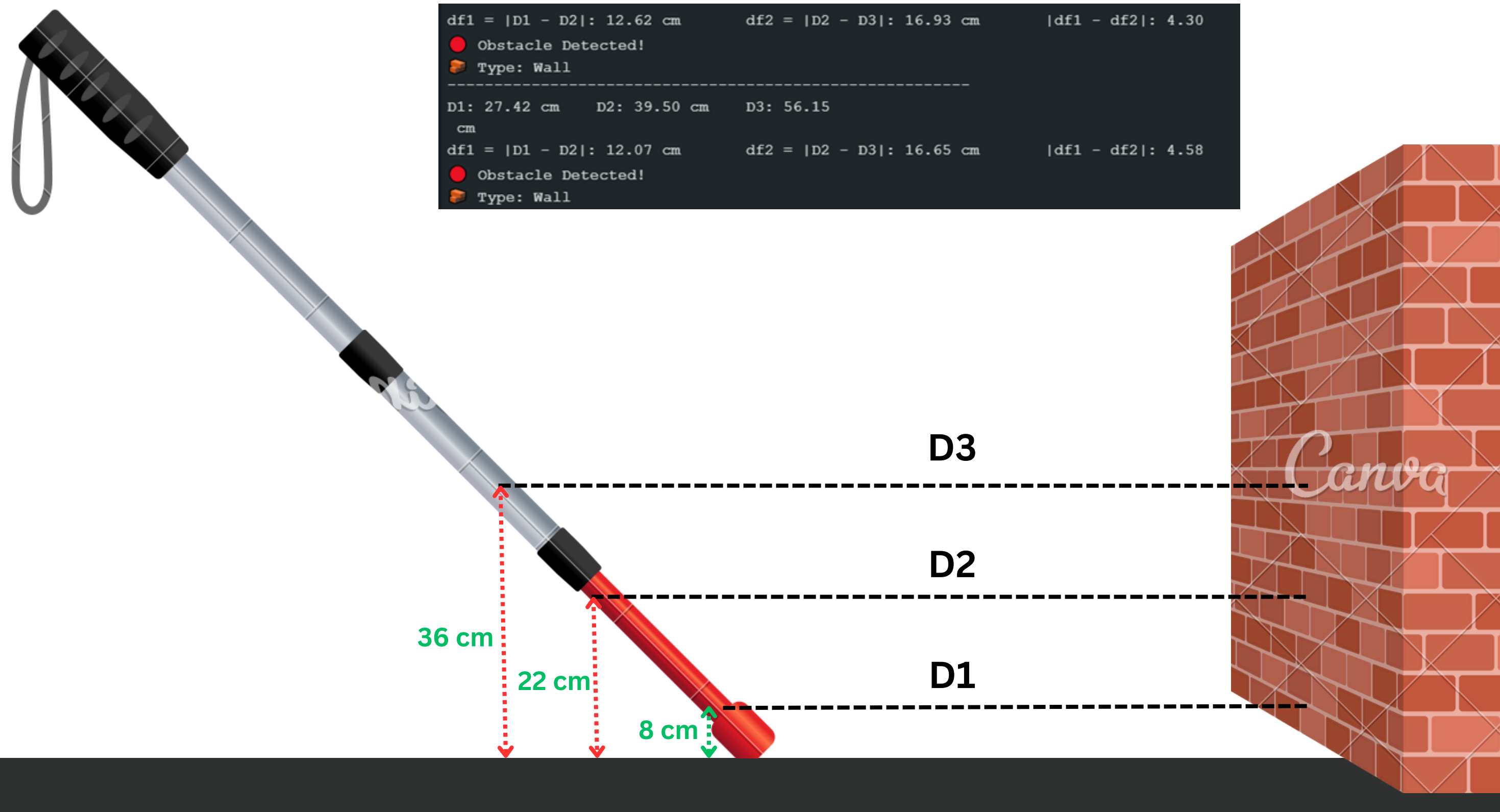


Wall Detection

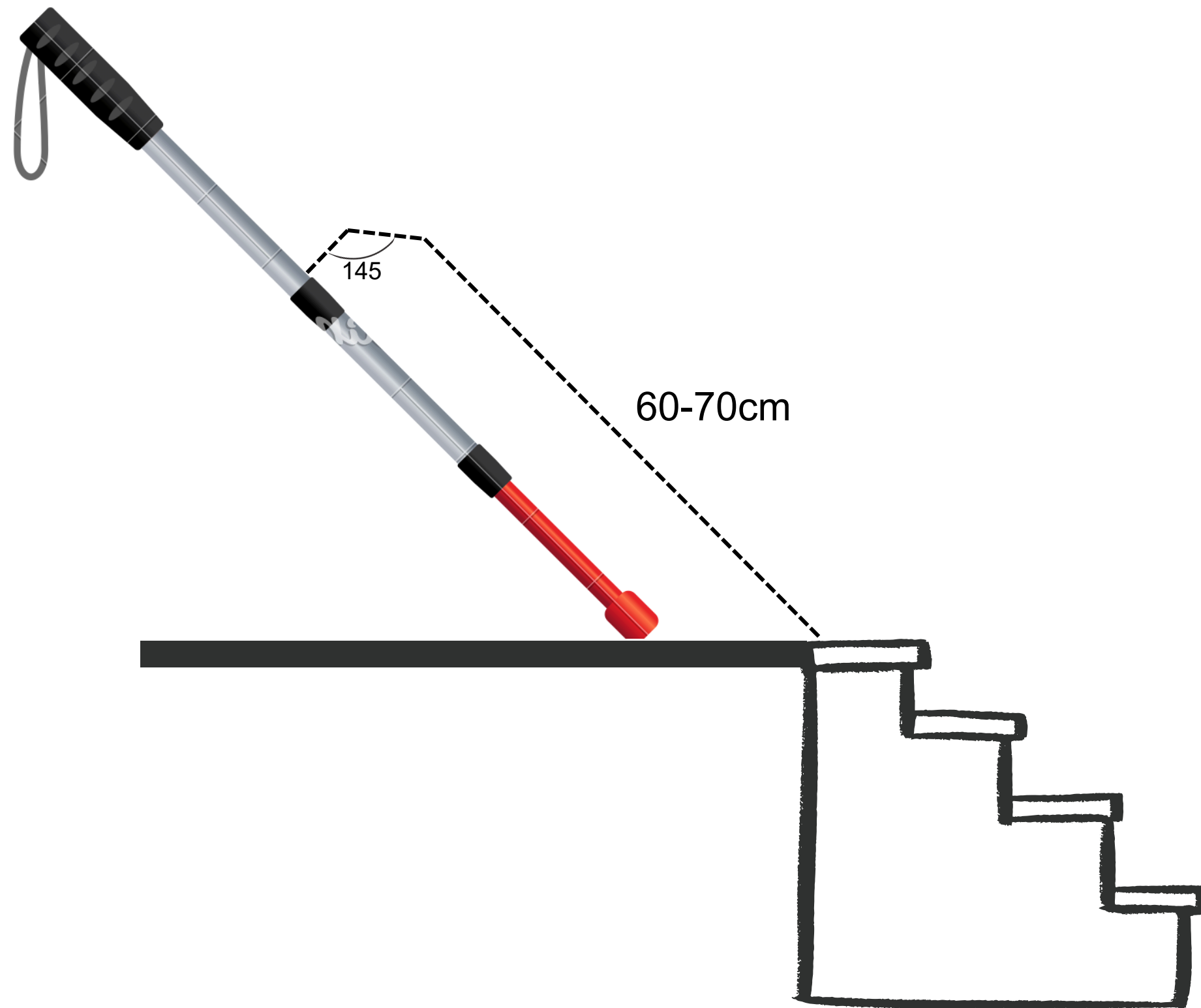
A wall is flat and vertical; all sensors will return similar distances.
This means $df1 \approx 0$, $df2 \approx 0$, and $\text{stairDiff} \approx 0$



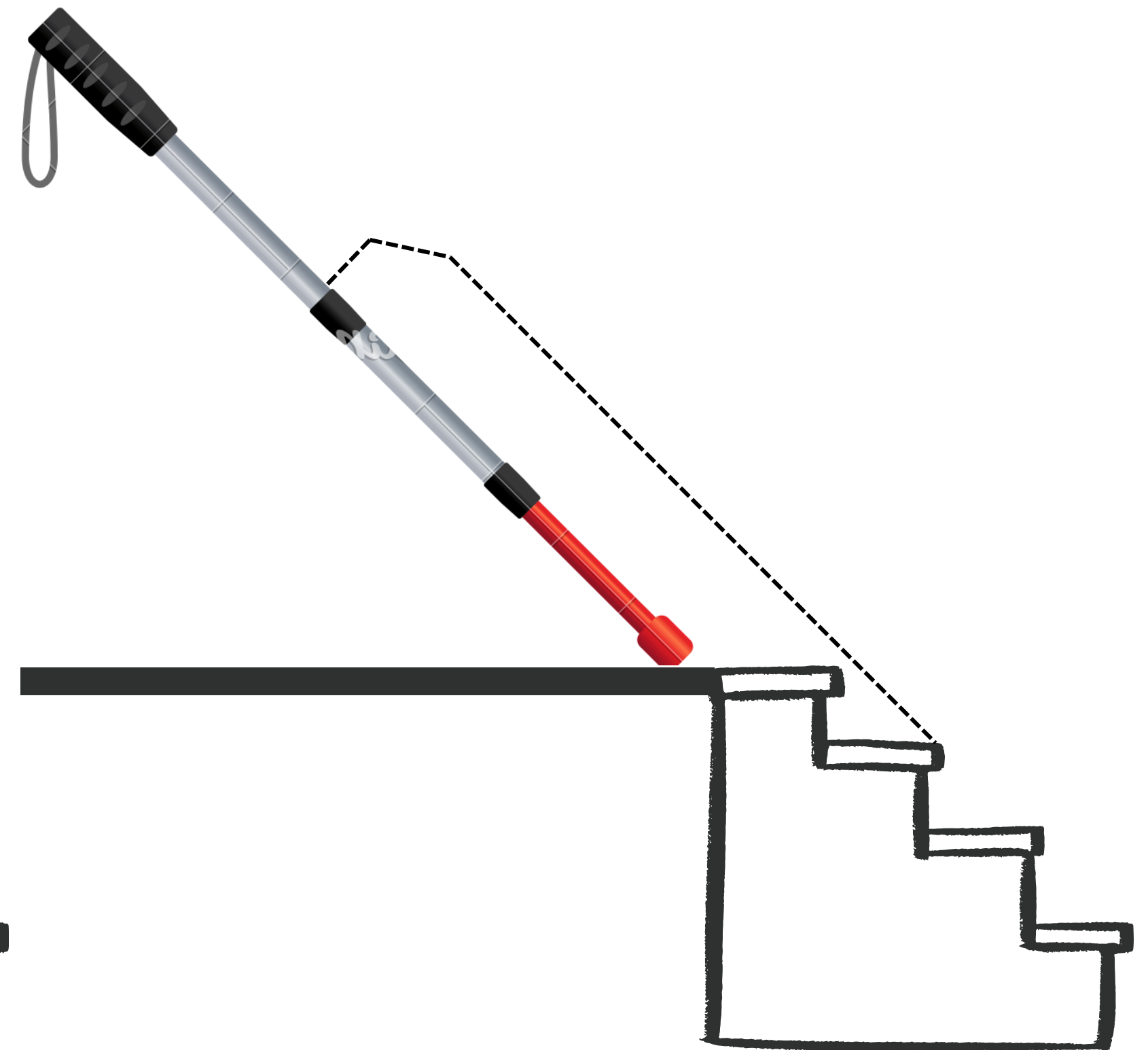
Wall Detection



DownStair Detection

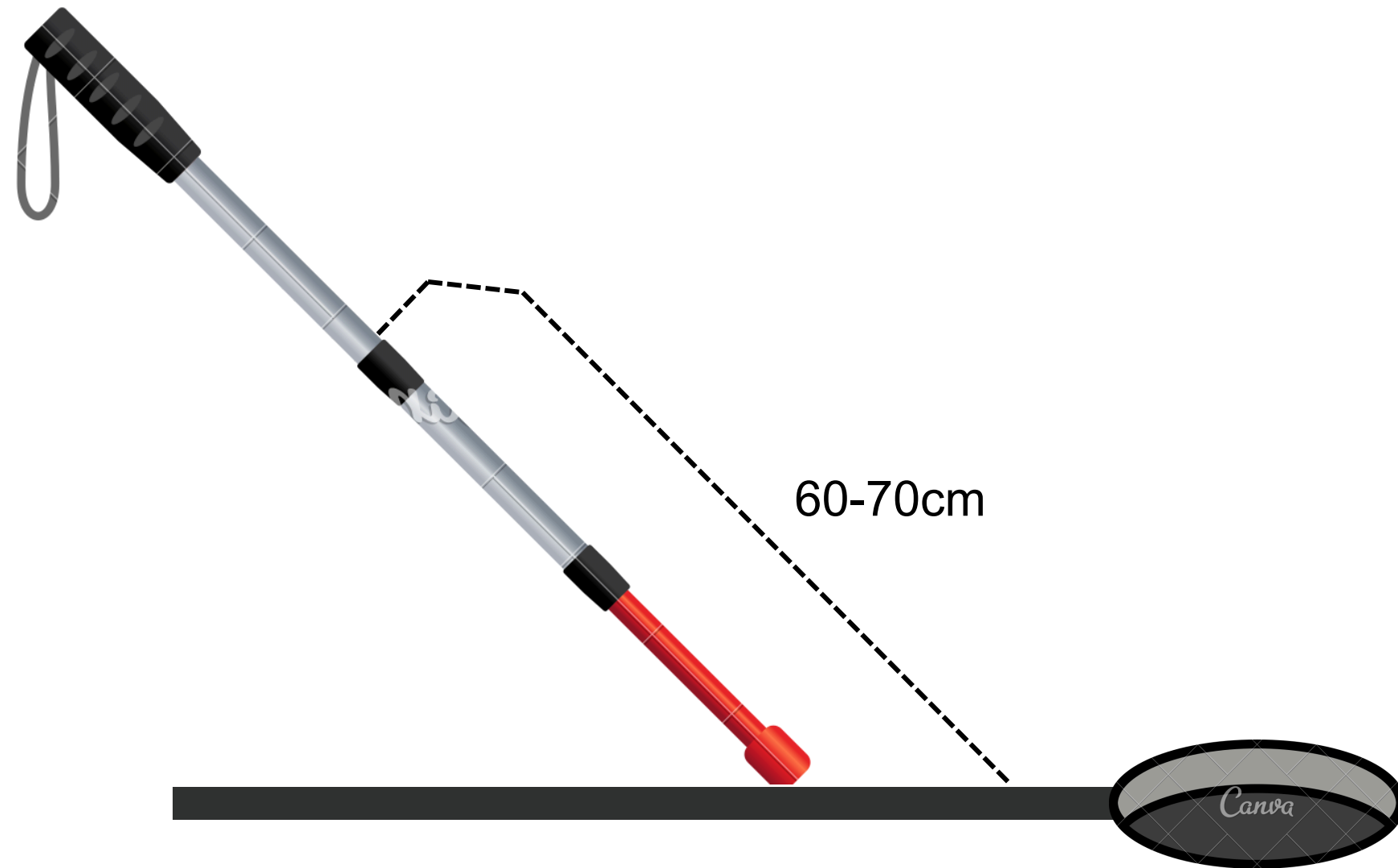


Before DownStair Detection

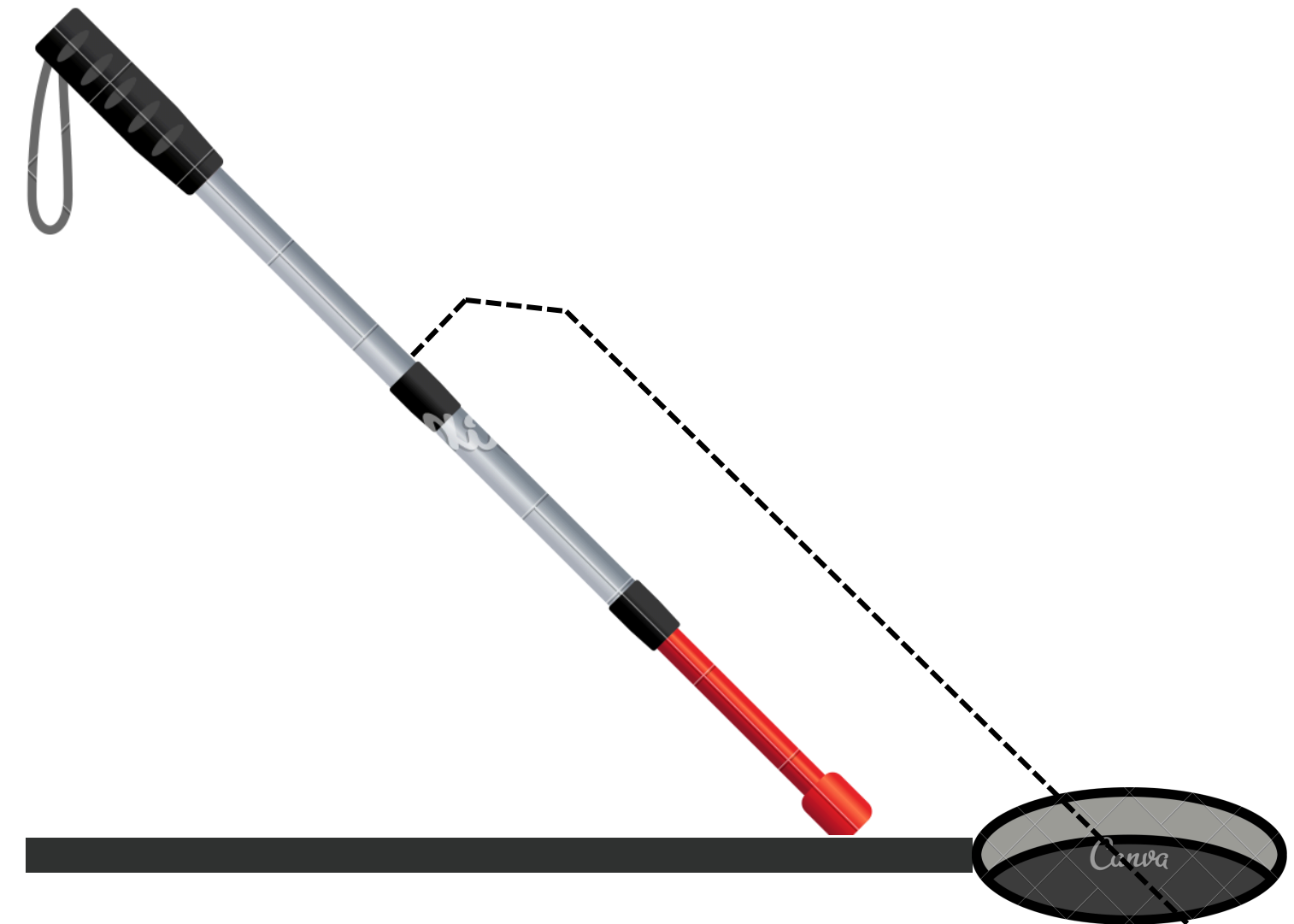


During DownStair Detection

Manhole Detection

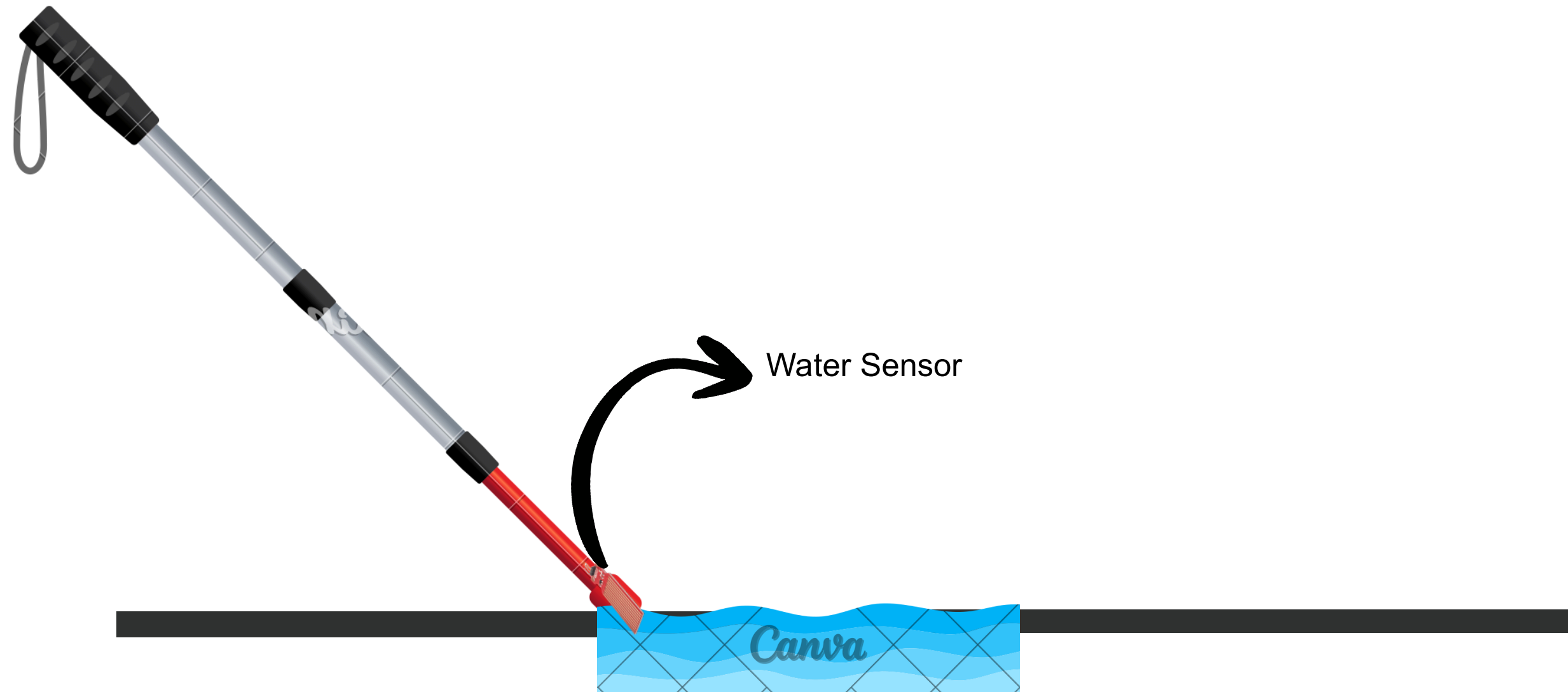


Before Manhole Detection

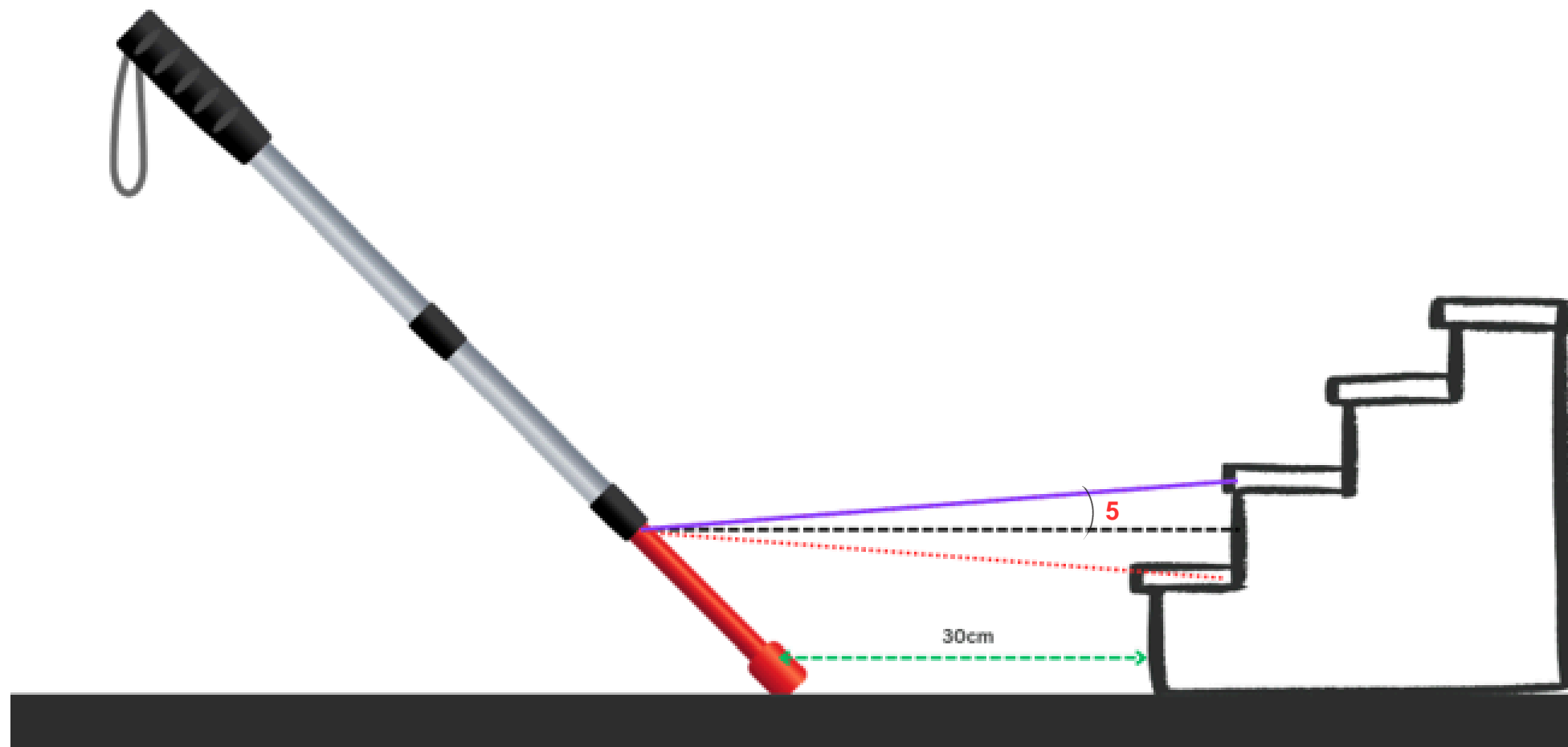


During Manhole Detection

Water Path Detection



Stick Angle Measurement



For S_1 ,

$$\sin \theta = \frac{7}{30} \approx 0.2333 \Rightarrow \theta \approx 13.5^\circ$$

For S_2 ,

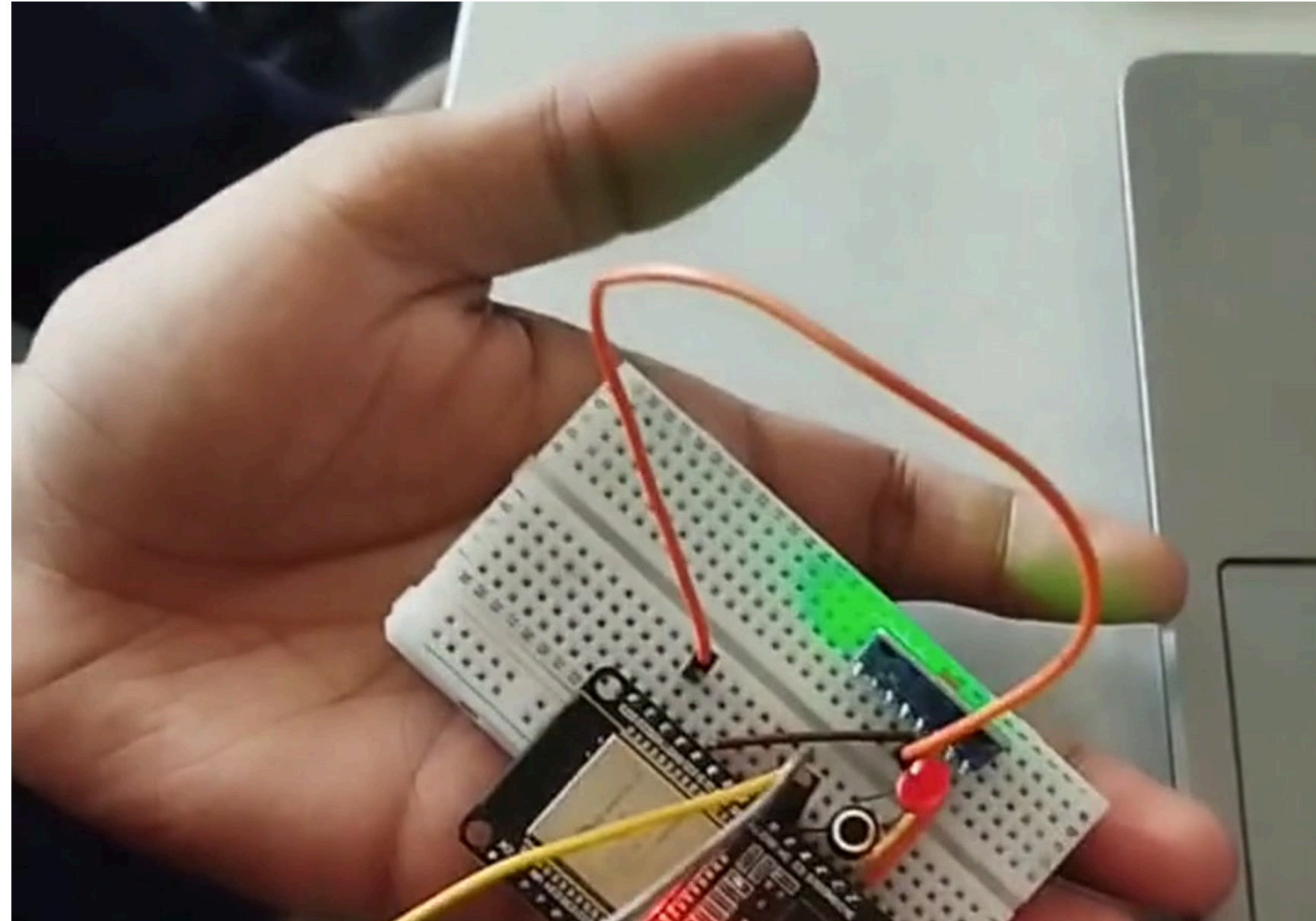
$$\sin \theta = \frac{7}{94} \approx 0.0745 \Rightarrow \theta \approx 4.26^\circ$$

For S_3 ,

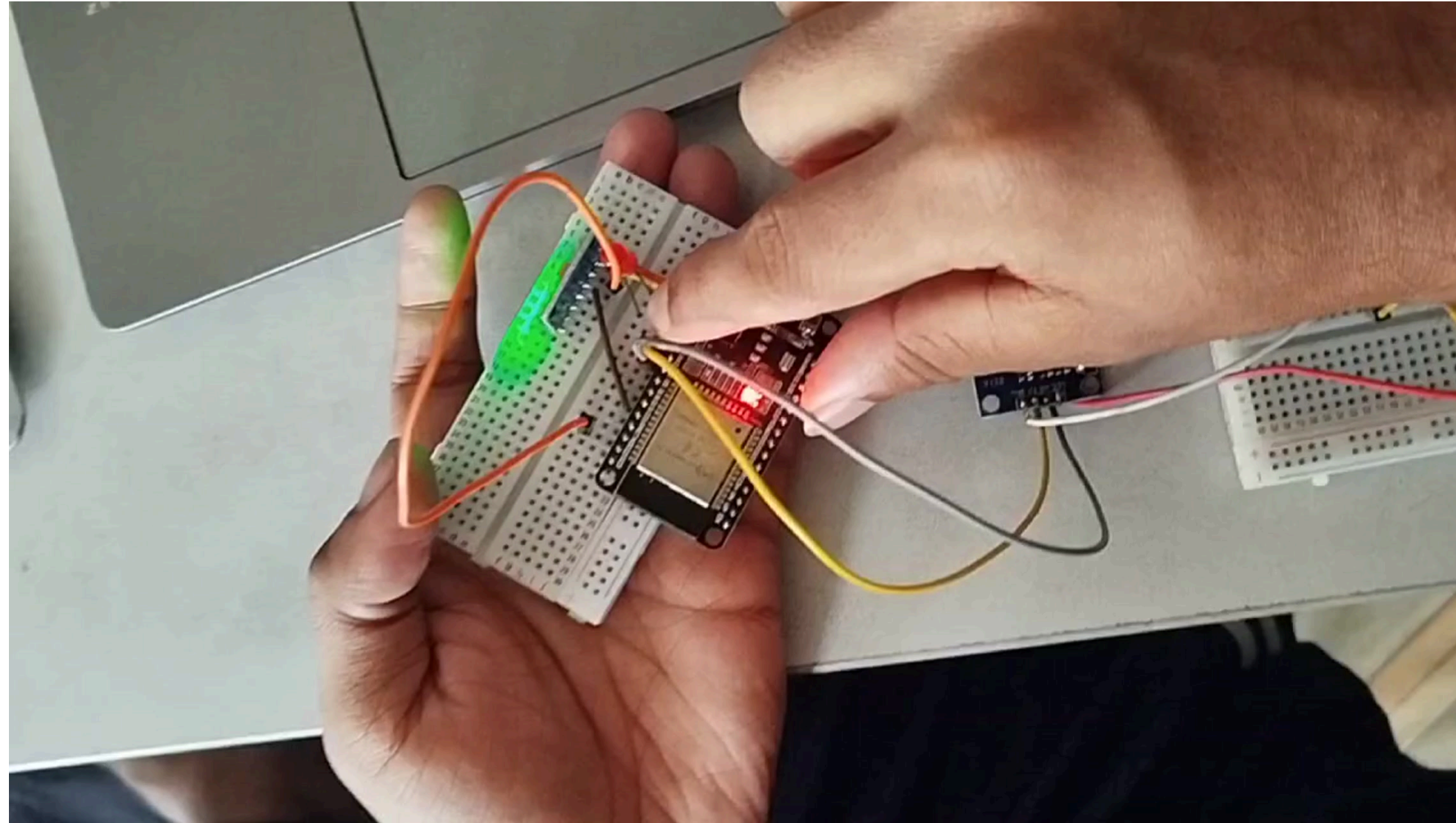
$$\sin \theta = \frac{7}{58} \approx 0.1207 \Rightarrow \theta \approx 6.93^\circ$$

Note: For more accurate results, we have considered the angle to be 5° .

Fall Detection Alert and Angle Change



Emergency Help Button



Angle Detection



Angle Is measured Using Accelerometer

Angles:
Angle X: 2.56°
Angle Y: 88.79°

GPS Implimentation

Continuously reads serial data from the GPS module.
Updates the latitude and longitude when a valid location fix is received

DD (decimal degrees)*

Latitude

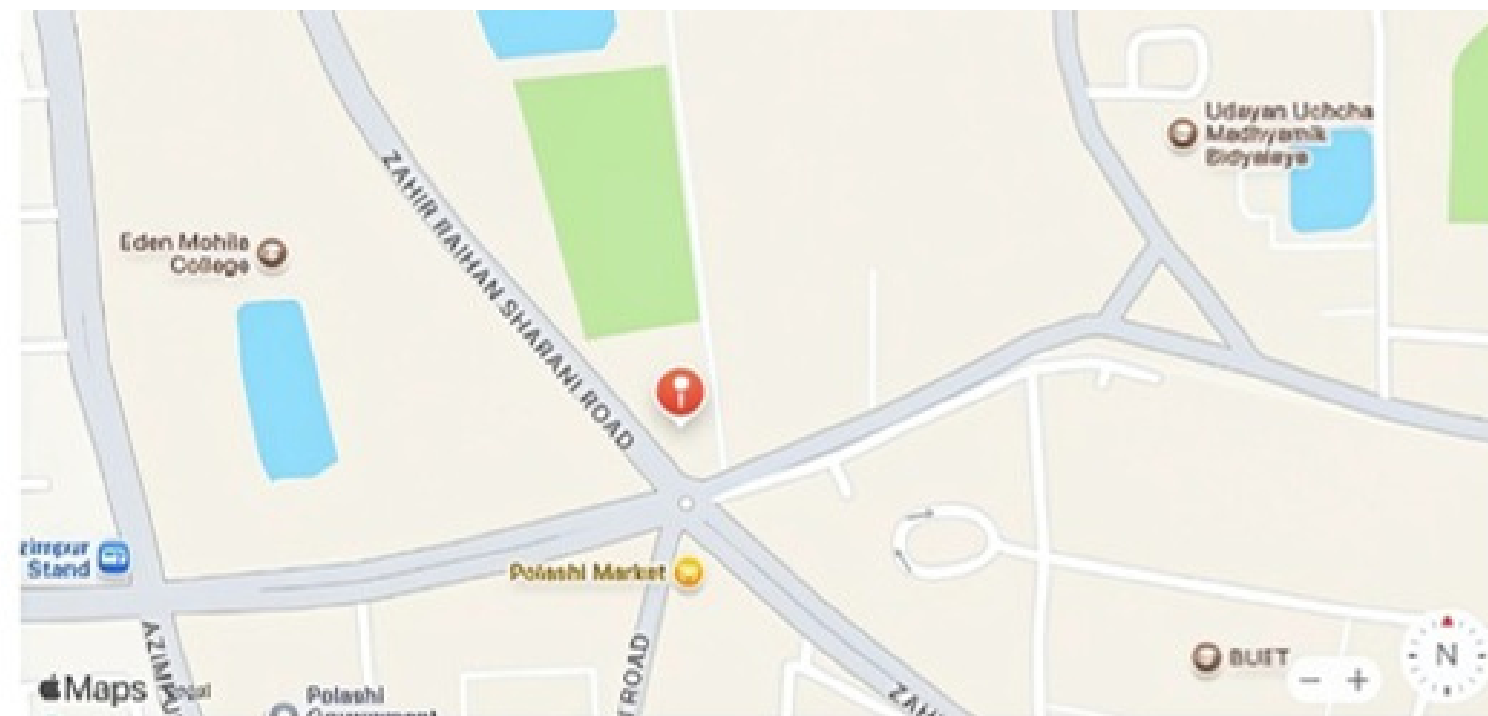
Longitude

Lat,Long

DMS (degrees, minutes, seconds)*

Latitude ☒ N ☐ S ° ' "

Longitude ☒ E ☐ W ° ' "



 **Location:**
Lat: 23.726336
Lng: 90.388541

Cost Analysis of Components

	Component	Low Range (BDT)
0	ESP32 Microcontroller	450
1	Arduino UNO	400
2	Accelerometer (MPU6050)	150
3	Ultrasonic Sensors (HC-SR04)	200
4	LDR (GL5528)	10
5	Battery Management Circuit (TP4056)	50
6	Li-ion or Li-polymer Battery (3.7V)	80
7	Buzzer	20
8	Walking Stick Frame	150
9	Wires & Connectors	150
10	Prototyping Board	100
11	Resistors, Capacitors, Transistors	50
12	Switches/Buttons	10
13	Total	1820



Division of Work Among Group Members

Working Sector	2006127	2006128	2006129	2006130
Sensor Integration (GPS, Ultrasonic, Force,	✓	✓		✓
Microcontroller Programming (Embedded C/C++)	✓	✓	✓	
WebApp Development	✓		✓	
Testing & Iteration (Prototype testing, user feedback)	✓	✓	✓	✓
Research & Documentation (Literature review, final report)	✓	✓	✓	✓



Timeline for Project Completion

Phase	Start Week	End Week
Phase 1: Research & Requirement Analysis	5	6
Phase 2: Hardware & Software Design	6	7
Phase 3: Prototype Development	7	8
Phase 4: Web App Development	8	9
Phase 5: Testing & Iteration	9	10
Phase 6: Final Deployment & Documentation	10	11



Course Outcomes (COs) and Program Outcomes (POs)

CO No.	CO Statement	PO(s)	Explanation
CO1	Utilize VerilogHDL, ARM assembly, C programming, and microcontrollers boards for implementing and analyzing theoretical concepts at software and hardware levels	PO(e)	The project involves the use of microcontrollers (ESP32) and programming (C/C++) to implement real-time data processing and sensor integration, fulfilling the hardware/software interaction requirement.
CO2	Design an embedded system with appropriate considerations to safety, cultural, societal, and environmental considerations	PO(c)	The smart walking stick design considers the safety of elderly and visually impaired individuals, environmental concerns, and social impact by improving mobility and reducing accidents.
CO3	Assess the impact of embedded systems project design on Societal, Health, Safety, Legal and Cultural Issues	PO(f)	The project directly addresses societal health issues such as falls, safety hazards, and the independence of elderly and visually impaired people, which is a major societal concern.
CO4	Evaluate Sustainability and the Impact of the Designed embedded systems Project in the Societal and Environmental Contexts	PO(g)	The smart walking stick contributes to social sustainability by enhancing the quality of life for elderly and visually impaired individuals, providing long-term mobility and safety benefits.
CO5	Demonstrate membership and leadership in designing embedded system related problem solving	PO(i)	This project requires collaboration in a team to integrate various sensors and technologies for the development of the smart walking stick. Leadership in design and problem-solving is essential for its success.
CO6	Communicate effectively on embedded system related design with presentation and detailed report	PO(j)	Clear communication is vital for documenting the design process, sharing technical specifications, and presenting the system to stakeholders, including caregivers and family members.
CO7	Demonstrate project management and cost analysis for embedded system projects	PO(k)	The development of the smart walking stick will require managing the project, analyzing costs, and ensuring that the final product is feasible, effective, and within budget.



References

Rahman, A. A., & Abdul-Rahman, A. (2022). A Smart Walking Stick for Elderly People – Part 2 : Design Configuration, Technical Analysis and Prototyping. Journal of Engineering and Science Research , 6 (6), 37–58.

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